LEVEL UP ACADEMY

Class 10 | GSEB | Science

ANSWER KEY & MARKING SCHEME

Ch 7 Test (GSEB pattern): How do Organisms Reproduce?

Chapter / Topic: How do Organisms Reproduce?

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SECTION A

Q1. Asexual reproduction takes place through budding in: **[1]**

Answer: (c) yeast

Q2. Multiple fission is seen in: **[1]**

Answer: (d) Plasmodium

Q3. Which of the following is NOT a part of the female reproductive system in human beings? **[1]**

Answer: (b) Vas deferens (part of the male system)

Q4. Which organism reproduces by fragmentation? **[1]**

Answer: (a) Spirogyra

Q5. The anther contains: **[1]**

Answer: (a) pollen grains

Q6. Which part of a flower develops into the fruit? **[1]**

Answer: (d) Ovary

Q7. Planaria can grow a complete new organism from a cut body piece. This is called: **[1]**

Answer: (c) regeneration

Q8. The transfer of pollen grains from the anther to the stigma is called: **[1]**

Answer: (b) pollination

Q9. Rhizopus (bread mould) reproduces by: **[1]**

Answer: (b) spore formation

SECTION B

Q10. What is the importance of DNA copying in reproduction? **[2]**

Answer: DNA carries the information for the body design. Copying DNA passes this information from parent to offspring, so the offspring are similar to the parent. Small errors in copying lead to variations, which help in survival over time.

- Passes on body design information: 1
- Leads to variation: 1

Q11. Give two advantages of sexual reproduction over asexual reproduction. **[2]**

Answer: (1) It combines DNA from two parents, so it creates more variation. (2) Variation increases the chance that some individuals survive if the environment changes, which helps the species survive.

- Each advantage: 1

Q12. Differentiate between self-pollination and cross-pollination. **[2]**

Answer: In self-pollination, pollen is transferred to the stigma of the same flower (or another flower on the same plant). In cross-pollination, pollen is transferred to the stigma of a flower on another plant of the same species, by agents like wind, water or animals.

- Each correct definition: 1

SECTION C

Q13. Answer the following about events after fertilisation in a flower: **[3]**

(i) It divides several times to form an embryo inside the ovule. [1]
(ii) The ovule develops into a seed, and the ovary develops into a fruit. [1]
(iii) They may shrivel and fall off. [1]

Q14. Answer the following about puberty: **[3]**

(i) Boys: testosterone. Girls: oestrogen. [1]
(ii) (Any two) Thick hair grows in the armpits and genital area; the skin becomes oily and pimples may appear; height and weight increase quickly. [1]
(iii) Testes (in boys) and ovaries (in girls). [1]

SECTION D

Q15. A gardener wanted many rose plants with exactly the same flower colour as one prized plant. He cut small pieces of its stem with buds and planted… **[4]**

(i) Cutting (vegetative propagation). [1]
(ii) Buds on the potato tuber (a modified underground stem). [1]
(iii) They are produced asexually, so they are genetically almost identical to the parent. Limitation: there is little variation, so all the plants may be equally affected by a disease or a change in the environment. [2]

- Asexual, genetically similar: 1
- Limitation: 1